20180312 TaMoS Multiple Choice Questions Natural Science

Vetenskapsteori och vetenskaplig metodik (naturvetenskap) (Kungliga Tekniska Högskolan)

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Final exam
Theory and Methodology of Science

Natural science

Course codes
AK2030/AK2036/AK2050/F1N5113/FAK3024

March 12, 2018

General information

The exam consists of three parts: a 'multiple choice' section (Part I) comprising 20 points, a 'two sentences' section (Part II) comprising 5 points, and a 'short answer' section (Part III) comprising 25 points. Pen/pencil, rubber and paper dictionary are the only allowed external materials.

Minimal passing score is 25 points of which at least 5 points must be awarded in Part III. Grades are awarded according to the table below (for master students), or according to the table on p. 2 (for PhD students).

Master students

Points	**Grade**
44.0–50.0	A
39.0–43.5	B
34.0–38.5	C
29.0–33.5 (of which $\geq$ 5 from Part III)	D
25.0–28.5 (of which $\geq$ 5 from Part III)	E
22.0–24.5 (of which $\geq$ 5 from Part III)	Fx
0.0–21.5 or <5 from Part III	F

Bonus points from the preparation and lecture participation questions will be added to the total sum if the exam score is 25 or above.

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PhD students

Points	**Grade**
35.0–50.0 (of which ≥ 5 from Part III)	Pass (without need for bonus points)
25.0–34.5 (of which ≥ 5 from Part III)	Pass if the student's bonus points bring the total sum to 35.0 or above; Extra assignment (FX) required if the student's bonus points bring the total sum to 29.0–34.5; Otherwise Fail
29.0–34.5 (of which ≥ 5 from Part III)	Extra assignment (FX) required (if the student's bonus points do not bring the total sum to 35.0 or above)
0.0–24.5 or <5 from Part III	Fail

The exam is four hours long. You are free to leave after one hour.

Examiner: Prof. Till Grüne-Yanoff, Division of Philosophy, KTH. Visiting address: Teknikringen 76, bottom floor.

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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives.

1. Does methodology require scientists to completely eliminate the risk of error in science? Select all correct answers.

   a) Yes. Committing errors shows that the related beliefs were unjustified.
   b) No. Science involves a trade-off between the utility of having knowledge, and the risk of being wrong.
   c) No. One argument for this is that science cannot eliminate all errors, only manage potential sources of error.
   d) Yes. Erroneous claims are unscientific.

2. What is the relationship between *scientific methods* and *methodology*? Select all true claims.

   a) Methodology provides empirical data for problems that methods cannot solve.
   b) Methodology is about method choice.
   c) For different methods, the same methodology might apply.
   d) Methodology is another word for a collection of methods.

3. Which of the following sentences can be used as hypotheses?

   a) "Do not add water to strong acids!"
   b) "The Ebola epidemic was caused by bacteria."
   c) "Computer program of type A uses more memory than computer program of type B."
   d) "Individuals with high level of education judge climate change as a high risk."

4. What is true about *inductive inferences*? Select all true claims. Inductive inferences ...

   a) ... are fallible.
   b) ... require justification.
   c) ... are truth-preserving.
   d) ... are explicative.

5. What is true of Karl Popper's falsificationism? Select all correct answers. The falsificationist...

   a) ... seeks to falsify hypotheses with observable evidence.
   b) ... considers only those hypotheses as true that have withstood many falsification attempts.
   c) ... conjectures (proposes) falsifiable hypotheses.
   d) ... never accepts any hypothesis as true – only maintains non-falsified hypotheses as so far not rejected.

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6. Which of the following claims about systematic error is correct? Select all true claims.

   a) Systematic error affects the degree to which repeated measurements under unchanged conditions show the same results.
   b) If the cause of a systematic error is identified, then the error can usually be corrected.
   c) Reducing systematic errors increases accuracy, but not precision.
   d) Systematic error can be reduced by repeating measurements and averaging over outcomes.

7. When one *ranks* movies on the scale 1,2,3,4,5, what kind of measurement scale is employed? Select all correct alternatives.

   a) An interval scale.
   b) The kind of scale that is also used when describing the final positions in a marathon.
   c) A scale that represents more information than an ordinal scale.
   d) A scale that represents less information than an interval scale.

8. What is correct about blinding in experiments? Select all correct alternatives.

   a) Experimental blinding is *not* an instance of experimental control by elimination.
   b) Double blinding requires that the experimenter is ignorant of which variables are dependent and which are independent.
   c) You could use double blinding to reduce observer expectation effects (i.e. the interpretation problem).
   d) One condition for single blinding is that subjects do not know which treatment they are getting.

9. Which of the following claims about experiments and observational studies are correct? Select all true claims.

   a) Experiments include control and manipulation.
   b) In observational studies one does not control all relevant variables, while in experiments one always successfully controls all relevant variables.
   c) In observational studies one manipulates only the variables that are thought to influence the outcome.
   d) Inferences from observational studies might be as good as those from experiments.

10. Identify the relevant epistemic virtues of scientific models from the following list.

    a) Robustness
    b) Non-computability
    c) Precision
    d) Simplicity

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11. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly? Select all correct alternatives.

    a) Some targets cannot be handled directly for practical reasons.
    b) We often do not know anything about the target.
    c) It is too expensive to handle some targets directly.
    d) Ethical or political reasons prohibit the direct manipulation of some targets.

12. Which of the following claims about statistics are true? Select all correct options.

    a) The application of standard statistical tools to good-quality data can lead to false conclusions.
    b) It is always clear from the descriptions of the statistical concepts and tools themselves when one should apply them.
    c) One can only "lie with statistics" by using invalid statistical methods, or by applying valid statistical methods to junk data.
    d) Statistics always lies.

13. Which claims are true about Bayesian statistics? Select all correct alternatives.

    a) Bayesian statistics can evaluate a large set of competing hypotheses.
    b) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.
    c) In Bayesian statistics, the probability of a hypothesis is changed if new evidence is observed.
    d) Bayes theorem decides whether to accept or reject a single hypothesis.

14. Which claims are true about the relation between correlation and causation? Select all correct claims.

    a) Correlation measures the association between variables. Causation measures the productive influence of one variable on another.
    b) Correlation between two variables is symmetric. The causal relation between two variables in asymmetric.
    c) That there is a common cause of two variables X and Y explains why there is a causal relation between X and Y, although there is no correlation between X and Y.
    d) Many different causal models between two variables are compatible with the observation that the two variables are correlated.

15. What is true of tacit knowledge?

    a) Teaching tacit knowledge typically requires personal contact and interaction.
    b) To know something tacitly means that one has a skill, experience or idea without necessarily being able to describe it explicitly.
    c) Tacit knowledge is taught mainly through engineering textbooks and manuals.
    d) The ability to ride a bicycle is an example of tacit knowledge.

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16. Which claims are true either of methodological individualism or holism? Select all true claims.

    a) Methodological holism defends the view that parts of a whole are in intimate interconnection, such that they cannot exist independently of the whole.
    b) Methodological individualism claims that there are no social entities in the world, only individuals (and families).
    c) Methodological holism is compatible with the claim that all social phenomena are produced by individual interactions.
    d) Methodological holism claims that there are good social explanations that do not provide individualist accounts.

17. Which of the following strategies can help improve the accuracy of interpretation of behavior in a usability experiment? Select all correct alternatives.

    a) Use inter-subjective checks.
    b) Make ad hoc modifications whenever suitable.
    c) Ensure that the belief-desire attributions are circular.
    d) Observing indications of cognitive processes, for example a "thinking out loud protocol".

18. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

    a) ... implies that if scientist A creates a new detergent, then scientist A must be able to show that the new detergent is not harmful.
    b) ... is of no concern in ideal democracies.
    c) ... imposes a moral constraint on researchers.
    d) ... states that when an activity raises threats of harm to human health or the environment, precautionary measures should be taken even if some cause and effect relationships are not fully established scientifically.

19. What is true about *utilitarianism*? Select all correct alternatives. According to utilitarians ...

    a) ... moral character is more important than moral actions.
    b) ... utility, often interpreted as happiness in some form, is the sole good.
    c) ... everything else equal, killing one person to save the lives of two others could be the right thing to do.
    d) ... individuals have rights, and there are some things we cannot do to others without violating these rights.

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20. A decision has to be made between four alternative courses of action.
On alternative A, there are two possible outcomes: 100 people dying (probability 0.6), or no one dying (probability 0.4).
On alternative B, there are four possible outcomes (equal probabilities): 20 people dying, 100 people dying, 400 people dying or 4 people dying.
On C, there are only two possible outcomes: 800 people dying (probability 0.01) or no one dies (probability 0.99)
On D, finally, either 30 people die, or no one dies (equal probabilities).
Select all correct statement(s), given the expected value criterion for decisions involving risks:

a) B is the fourth best alternative.
b) C is the best alternative.
c) A is the best alternative.
d) D is the best alternative.

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PART II – Two-sentences questions

Instructions for two-sentences questions: Write one or two sentences for each of the questions below on a separate sheet of paper, *not* on the exam. You can use a single sheet of paper for all five questions. One point will be awarded for each correct answer. A partially correct answer will be awarded half a point. An incorrect answer, or an omitted answer, gives no points. Writing more than two sentences is not needed, but will not lead to a deduction of points. However, an answer that lacks focus indicates that you have not fully understood the concepts and therefore might result in a point reduction.

1. Name the three main frameworks of normative ethics.
2. Describe convergent validity and divergent validity.
3. What is the distinction between *explanans* and *explanandum*?
4. What makes a model robust with respect to some result?
5. What does it mean to say that an experiment is repeatable?

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PART III – Short answer questions

The following questions should be answered on separate sheets of paper. *Hints on answer lengths*: If you answer a question worth 5 points with a single sentence, it is unlikely that your answer is elaborate enough to be awarded 5 points. On the other hand, if your answer is informative and to the point, there should be little use for more than one page for a 5 points question and two pages for a 10 points question.

1. Describe Mill's method of difference, and show how it works with an example. Explain what possible sources of error there are when using the method and discuss how these would affect the conclusion in your example. (5p)

2. What are the main differences between inductive and deductive inferences? Furthermore, name and describe four types of scientific inferences and categorize them as being deductive or inductive. Provide examples of each type of scientific inference. (5p)

3. What does it mean that indirect observation is theory dependent? Explain how this is important for the measurement process, and discuss how nomic measurement and theory dependence can be seen as an instance of the Duhem-Quine-problem. (5p)

4. A researcher has formulated the following hypothesis: "The lead content of agricultural soil has a negative effect on the crop yield of corn."

   (a) Describe how you would design a field experiment to test this claim. (3p)

   (b) Would it be possible to investigate the hypothesis with a natural experiment instead? Explain how! What conditions would have to be in place? (2p)

   (c) What kind of study do you think is best to test the hypothesis: natural experiment, field experiment, or laboratory experiment? Compare the different types and motivate your preferred kind of study! (5p)

   For all questions (a)-(c) make sure that you give concrete examples, use the right terms and concepts from the course and define the terms you use!

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TaMoS: Answer sheet for mcq's

Student's name:

Personal number:

	a)	b)	c)	d)
1.				
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	a)	b)	c)	d)
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